

A Hierarchical Meta-Learning Driven Adaptive Brain Region Flow Fusion Method for EEG-Based Emotion Recognition[☆]

Hao Liu^{a, #}, Yindong Dong^{a, #, *}, Yiming Tang^b, Jingjing Deng^a, Yong Ye^a, Yubo Xing^a, Xiule Wang^a

^a School of Artificial Intelligence, Anhui Agricultural University, Hefei, China

^b School of Computer Science and Information Engineering, Hefei University of Technology

ARTICLE INFO

Keywords:

adaptive brain region flow
EEG
emotion recognition
feature fusion
meta-learning

ABSTRACT

EEG (electroencephalography)-based emotion recognition has garnered substantial attention in affective computing. However, existing state-of-the-art methods still face two major challenges: 1) It is difficult to strike a balance between a priori physiological consistency and data-driven flexibility in the functional parcellation of brain regions; 2) Ineffective synergistic adaptation to individual differences and local-global feature correlations. To tackle these, we propose HMABRF, a hierarchical meta-learning-driven adaptive brain region flow fusion method for EEG-based emotion recognition. Guided by affective neuroscience priors, we design an adaptive brain region flow partitioning mechanism. We begin by using self-supervised contrastive learning to enhance the semantic consistency of channel features, and then employ a differentiable Gumbel-Softmax function to perform dynamic clustering of sub-flows; in addition, based on a hierarchical meta-learning framework, we develop LTIM (Local Tuned Interaction Module) and GTCM (Global Temporal Calibration Module). LTIM enhances local multi-scale feature interaction through its built-in Bilateral Feature Interaction and Enhancement (BIE), specifically adapting to individual differences in local details; GTCM leverages Mamba and the Contextual Temporal Interaction Relay (CIR) to capture long-range temporal dependencies and accurately adapt to subject-specific global long-term feature patterns across the entire sequence, while using a self-supervised alignment loss to enhance the semantic consistency between local and global features. The experimental results show that HMABRF outperforms existing state-of-the-art methods on the SEED and SEED-IV datasets.

1. INTRODUCTION

As a core driver of human cognition and behavior, the accurate recognition of emotions holds significant practical value in fields such as human-computer interaction (Zhang, Li and Wang, 2023), (Du et al., 2023), mental health monitoring (Li, Zhang, Tiwari and Li, 2023), (Liu et al., 2018), and intelligent education (Alqahtani, Katsigiannis and Ramzan, 2021), (Alarcão and Fonseca, 2019). Compared to behavioral cues such as facial expressions (Sukumar et al., 2024) and speech (Zhu, Dai, Hu and Li, 2020), which are easily masked by subjective factors, and peripheral physiological signals such as electrocardiograms (ECG) (Nita, Bitam, Heidet and Mellouk, 2022) and electrodermal activity (EDA) (Ariel D'Amelio et al., 2025), which only indirectly reflect

emotional states, electroencephalography (EEG) has become an ideal physiological signal source for emotion recognition due to its non-invasive nature, high temporal resolution, and ability to directly reflect neural activity in the brain (Apicella et al., 2022, Ding et al., 2023, Lim, Lew and Ang, 2024, Zhu et al., 2025, Yu, Li, Zang and Liu, 2023, Li et al., 2026). In recent years, researchers have conducted extensive studies in areas such as brain region delineation, individual adaptation mechanisms, and the integration of local and global features. For example, regarding brain region delineation, research by Professor Lü Baoliang's team (Zheng, Zhu and Lu, 2019), based on the SEED dataset (Zheng and Lu, 2015), (Duan, Zhu and Lu, 2013), identified key brain regions associated with different emotions—such as the bilateral temporal lobes and prefrontal cortex—laying the foundation for

[☆] Peer review under the responsibility of KeAi Communications Co., Ltd

* Corresponding Author.

E-mail addresses: liuhao666@stu.ahau.edu.cn (H. Liu), dongyindong@ahau.edu.cn (Y. Dong), tym608@163.com (Y. Tang), 1198659872@qq.com (J. Deng), yeyong@ahau.edu.cn (Y. Ye), st_xyb025@163.com (Y. Xing), mingmingxl26@163.com (X. Wang).

stands for two author contribute equally

<https://doi.org/10.1016/j.ijcce.2026.04.005>

Received 26 March 2026; Received in revised form 25 April 2026; Accepted 27 April 2026

Available online 28 April 2026

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physiologically-guided functional localization of brain regions; The AM-GCN model proposed by Zhou et al. (Zhou, Song, Gu and Ji, 2025) uses adaptive graph learning to construct data-driven dynamic brain region connections via GCN modules, eliminating the need for a fixed physiological topology and enabling dynamic modeling of brain region interaction patterns; Regarding individual adaptation mechanisms, Chen et al. (Chen, Fang, Yang and Zhou, 2025) employed a meta-learning framework to perform fine-tuning with few samples, thereby achieving effective adaptation to inter-subject differences; Zhou et al. (Zhou et al., 2025) designed the EEGMatch semi-supervised transfer learning architecture, which mitigates the issue of individual heterogeneity through multi-domain distribution alignment; In the field of feature fusion, the EEG-CSANet proposed by Cai et al. (Cai et al., 2026) employs a multi-branch parallel architecture and combines it with centralized sparse attention to achieve synergy between local details and global trends; Liu et al. (Zhou, Huang, Peng and Yin, 2025) proposed the miMamba model based on the Mamba architecture, which uses a multi-scale inverted structure to simultaneously capture both local details and global temporal dependencies in EEG, thereby significantly improving the efficiency of long-sequence modeling and recognition performance. They also validated the suitability and advantages of the Mamba architecture for EEG emotion recognition tasks; A review study (BioArt Editorial Office 2025) confirmed that the BiLSTM-CNN hybrid model can effectively capture both the short-term local responses and long-term global trends in EEG, achieving a recognition accuracy of 90.54% on the GAMEEMO dataset.

Currently, mainstream deep learning frameworks for EEG-based emotion recognition have evolved into diverse technical approaches, all of which serve as baseline methods for comparison in the experiments described in this paper: As classic convolutional architectures, DeepConvNet (Schirrmester et al., 2017) and EEGNet (Lawhern et al., 2018) perform spatiotemporal feature extraction from EEG data through deep convolutions and lightweight design, respectively; however, they lack the ability to adaptively segment brain regions and model long-term temporal sequences; TSception (Ding et al., 2023) focuses on modeling temporal dynamics and spatial asymmetry in EEG and does not account for individual differences; Although DGAT (Ding et al., 2025) uses a dynamic graph attention network to construct adaptive connections between brain regions, it remains constrained by fixed physiological prior constraints; AMDet (Xu et al., 2024) enhances global feature representation using a multi-dimensional attention Transformer, but fails to achieve coordinated adaptation between local and global features. Although the aforementioned methods have improved recognition performance in areas such as convolutional neural networks, graph neural networks, Transformers, and transfer learning, they have yet to resolve two major challenges: First, it is difficult to strike a balance between physiological a priori consistency and data-driven adaptability when delineating functional brain regions; traditional a priori delineations lack flexibility, while purely data-driven methods tend to deviate from neuroscientific logic; Second, existing models struggle to effectively coordinate the adaptation to individual differences with local-global features. Most current models treat individual adaptation and feature fusion as separate modules, which disconnects the relationship between local short-term features and global long-term trends, thereby limiting their generalization performance across subjects.

To address the aforementioned issues, we propose a method for emotion recognition called Hierarchical Meta-Learning-Driven Adaptive Brain Region Flow Fusion (HMABRF), which utilizes electroencephalography (EEG). The term “hierarchical” specifically refers to the hierarchical meta-learning framework designed in this paper; the detailed architecture will be described in the Methods section. This method integrates an adaptive dynamic brain region segmentation mechanism, a Local Tuned Interaction Module (LTIM), and a Global Temporal Calibration Module (GTCM). It strikes a balance between physiological a priori consistency and data-driven adaptability in brain region segmentation, while also enabling the coordinated adaptation of individual

variations and the correlation of local-global features. The main contributions of this paper can be summarized as follows: (1) We address the issue of over-reliance on either physiological a priori assumptions or data-driven methods in functional brain region segmentation by employing adaptive brain region segmentation. (2) We have innovatively designed a dual-module architecture comprising LTIM and GTCM, driven by hierarchical meta-learning, which enables the simultaneous modeling of local and global features while accommodating inter-subject variability, thereby overcoming the limitations of existing models. (3) We propose a hierarchical adaptive feature fusion strategy that encompasses both intra-substream and cross-substream feature aggregation, enabling efficient integration of whole-brain emotional features and improving the feature fusion mechanism for EEG emotion recognition. (4) The HMABRF method we propose achieves superior performance on both datasets, outperforming existing state-of-the-art methods.

2. DATASETS

We use the SEED (Zheng and Lu, 2015), (Duan, Zhu and Lu, 2013) and SEED-IV (Zheng et al., 2019) datasets to evaluate the model's performance. The SEED dataset contains EEG data from 15 Chinese participants, divided into three experimental phases (with approximately one week between each phase), with 15 trials in each phase. The EEG from each trial correspond to one of three emotional labels: positive, neutral, or negative; The SEED-IV dataset also contains EEG data from 15 subjects, each of whom completed three experimental sessions on different dates with 24 trials in each session. The EEG of each trial are assigned to one of the four emotional labels: happy, neutral, sad, and fear.

3. METHODS

The architecture of HMABRF is illustrated in Fig. 1, and this chapter elaborates on the specific composition and functional implementation details of each module in detail.

3.1. Input Preprocessing

- 1) Signal Structure Transformation and Time-Frequency Feature Extraction: First, the raw EEG data is processed using an overlapping sliding window strategy. Based on the effective duration of emotional signals in the SEED and SEED-IV datasets, a window length of 2 seconds and an overlap rate of 50% are selected for segmentation. After segmentation, each sample is represented as a two-dimensional tensor, which preserves the temporal integrity of the emotional neural responses while increasing the sample size. All samples ultimately form the input dataset for subsequent time-frequency feature extraction; After calculation and dimensionality reduction, the number of time-series subwindows in this experiment is set to $T' = 30$. The time-domain signal is then converted into a two-dimensional time-frequency matrix using the short-time Fourier transform (STFT); Next, we extract DE features across the five core frequency bands: delta (1–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–45 Hz), ultimately converting each sample into a three-dimensional feature tensor $X_i \in R^{C \times F \times T'}$ (Zhou, Tang and Yue, 2026).
- 2) Adaptive Brain Region Flow Partitioning: First, the 62 EEG channels are divided into three macro-brain regions based on theories of affective neuroscience, thereby establishing the biological basis for this classification. The prefrontal cortex (Etkin, Egner and Kalisch, 2011), (Gil, Seo and Lee, 2008) comprises 18 channels, including Fp1 and Fp2. Its core functions include emotional regulation, decision-making, and working memory, making it the primary brain region for emotional processing; The temporal lobe (Jha, Suvvari and Kumar, 2024) comprises 16 channels, including T7 and T8, and is

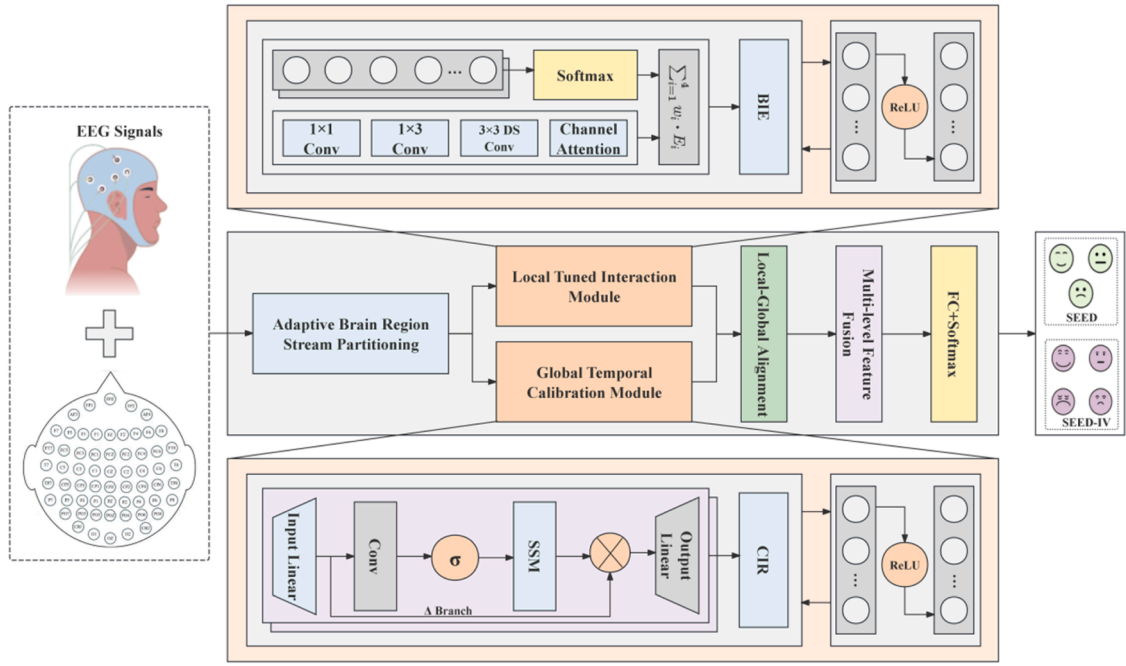


Fig. 1. Schematic Diagram of the HMABRF Architecture. Abbreviations used in the diagram include BIE (Bilateral Feature Interaction and Enhancement), CIR (Contextual Temporal Interaction Relay), and SSM (Structured State Space Model). Key modules are clarified as follows: Adaptive Brain Region Stream Partitioning denotes the adaptive brain region stream partitioning; LTIM (Local Tuned Interaction Module) is designed for local multi-scale feature capture; GTCM (Global Temporal Calibration Module) is responsible for global temporal feature modeling; Local-Global Alignment refers to the local-global feature alignment; Multi-level Feature Fusion represents the multi-level feature fusion, which includes intra-stream and inter-stream fusion steps.

primarily responsible for emotional memory encoding, auditory information processing, and the integration of emotional and semantic information; The parietal-occipital region (Li et al., 2022), (Tseng et al., 2024) comprises 28 channels, including P3 and O1, and focuses on the sensory representation of multimodal perceptual integration, spatial attention allocation, and emotional experience; Next, within each macro-flow, we construct a channel-specific embedder consisting of a 1×1 convolution and LayerNorm to transform the single-channel spatio-temporal features into 128-dimensional semantic vectors, thereby preserving the neural response specificity of different channels; We then introduce self-supervised contrastive learning by randomly sampling all 62 EEG channels within each training batch. Channels within the same macro-regional stream were paired as positive sample pairs, while channels from different macro-regional streams are paired as negative sample pairs. The ratio of positive to negative sample pairs within each batch is controlled at 1:3 to ensure training balance. The batch sizes for the contrastive learning branch are kept consistent with those of the main model branch, with 32 for the SEED dataset and 64 for the SEED-IV dataset. The temperature coefficient τ of the contrastive loss (InfoNCE) is optimized to 0.5 via grid search, effectively enhancing the semantic consistency of features belonging to the same functional flow channel through the contrastive loss; Finally, we perform clustering using the differentiable Gumbel-Softmax model, with the candidate range for the number of sub-streams N set to (Du et al., 2023, Zhu, Dai, Hu and Li, 2020). For each candidate value of N within this range, we calculate the information entropy of the corresponding feature distribution, as shown in Equation (1):

$$H(N) = - \sum_{i=1}^N p_i \log p_i \quad (1)$$

Where p_i denotes the distribution probability of features in the i -th

sub-flow. The information entropy $H(N)$ reflects the uncertainty of the feature distribution within sub-flows; the smaller $H(N)$ is, the stronger the semantic consistency of features within sub-flows becomes, and it can simultaneously balance the issues of computational redundancy and feature underfitting. Therefore, the value of N corresponding to the minimum value of $H(N)$ is selected as the optimal number of sub-streams. The final experimental results indicate that the optimal number of sub-streams for the prefrontal cortex is 5, for the temporal lobe is 4, and for the parietal-occipital lobe is 5. Based on these findings, an adaptive set of brain region streams, denoted as F , is formed. This set F aligns with the distribution characteristics of individual EEG data while remaining consistent with neurophysiological functional logic.

3.2. Local Tuned Interaction Module

The core objective of the LTIM is to accurately capture emotion-related local features, enhance feature interaction and complementarity, and dynamically adapt to individual variations in local details. It integrates the Mixture of Experts (MoE), Bilateral Feature Interaction and Enhancement (BIE), and a local meta-learner.

1). Mixture of Experts (MoE): To address the need for multi-scale differentiation of local emotional features in EEG, this paper designs a set of specialized expert networks, each tailored to model four core local feature dimensions. Expert1 uses 1×1 convolutions to capture inter-band correlations within a single temporal subwindow; Expert2 employs 1×3 convolutions to capture short-term temporal dependencies across consecutive temporal subwindows; Expert3 utilizes 3×3 depthwise separable convolutions (Wanzeng, Min and Jianhai, 2024) to extract local interactive features across channels and time; Expert4 introduces a channel attention mechanism to highlight emotion-related local responses in key channels within a substream. Therefore, this paper employs four expert networks to comprehensively cover the four core local feature dimensions mentioned above. This configuration was not chosen arbitrarily;

subsequent ablation experiments have confirmed that it represents the optimal balance between feature extraction capability and computational efficiency. This is combined with a gated network consisting of two fully connected layers and a Softmax activation function, which outputs the weights of the four experts and dynamically fuses them using the formula shown in Equation (2):

$$F_{moe} = \sum_{i=1}^4 \omega_i \cdot \text{Expert}_i(X_{sub}) \quad (2)$$

In this context, X_{sub} represents the sub-stream input features; this design enables the gate to precisely match the appropriate expert based on local feature patterns, thereby enhancing the specificity of feature extraction.

2)Bilateral Feature Interaction and Enhancement (BIE): First, the F_{moe} is linearly projected onto 128-dimensional query vector Q , key vector K , and value vector V . A sinusoidal temporal position encoding $PE(t)$ is then added to Q/K to preserve temporal information, as shown in Equations (3) and (4):

$$PE(t, 2k) = \sin(t/10000^{2k/d}) \quad (3)$$

$$PE(t, 2k + 1) = \cos(t/10000^{2k/d}) \quad (4)$$

Here, $d=128$ represents the feature dimension, and k is the dimension index. After computing the interaction features via scaled dot-product attention, the original local features are obtained through residual connections and LayerNorm to enhance training stability, as shown in Equation (5):

$$F_{local_raw} = \text{LayerNorm}\left(F_{moe} + \text{softmax}\left(\frac{(Q + PE)(K + PE)^T}{\sqrt{d}}\right)V\right) \quad (5)$$

3)Local Meta-Learner: To account for variations in local details across individual instances, we construct a parameterized meta-learner consisting of two fully connected layers and ReLU layers. This meta-learner adjusts the parameter adjustment amounts $\Delta\theta = [\Delta\theta_{moe}, \Delta\theta_{bie}]$ for the two modules based on the raw local features output by MoE and BIE, and updates the module parameters according to the following formula:

$$\theta_{update} = \theta_{init} + \alpha \cdot \Delta\theta \quad (6)$$

Here, θ_{init} is the initial parameter for population training, and the adaptation coefficient $\alpha=0.15$ is determined by grid search. This local meta-learner is a core component of hierarchical meta-learning. Its function is to generate personalized parameter updates based on differences in the distribution of local EEG features across different subjects, thereby preventing performance degradation in group-trained models during cross-subject testing. Ultimately, it outputs the adapted sub-flow local features $F_{local} \in R^{128}$, effectively addressing the challenge of individual heterogeneity in EEG emotion recognition.

3.3. Global Temporal Calibration Module

The core objective of GTCM is to capture long-term dynamic trends in the evolution of sentiment. It integrates the Mamba global feature extractor, the Contextual Temporal Interaction Relay (CIR), and the global meta-learner, forming a closed-loop local-global coordination

mechanism with LTIM.

1)Mamba Global Feature Extractor: Regarding the task of temporal modeling for EEG emotion recognition, current mainstream methods all have significant limitations: BiLSTM is prone to vanishing gradients, making it difficult to model long-range dependencies, while Transformers suffer from high computational costs due to their $O(n^2)$ complexity and are susceptible to noise interference. To this end, this paper employs Mamba as the core of global feature extraction. This efficient state-space model has $O(L)$ linear complexity and is well-suited for modeling long sequences of EEG time-series features. First, the sub-stream features output by the Adaptive Brain Region Flow are flattened into a $T \times 128$ temporal sequence, which is then fed into a two-layer Mamba module with a hidden layer dimension of 256 and with an expansion factor of 2. Leveraging Mamba's input-correlated selective propagation mechanism, this process dynamically filters out temporal redundant noise, prioritizing the retention of long-term trend features strongly correlated with emotional states, and outputs the initial global feature $F_{mamba} \in R^{128}$. Although the current sequence length of 30 is considered a medium scale, Mamba still offers significant advantages over standard self-attention mechanisms; its linear computational complexity can substantially reduce the cumulative computational overhead in models with multi-branch and multi-substream architectures. At the same time, the selective state space mechanism effectively filters out the vast amount of temporal redundancy and noise in EEG, avoiding the global information dilution issue inherent in standard self-attention mechanisms. Furthermore, its independence from Softmax normalization effectively mitigates the vanishing gradient problem during temporal modeling. Combined with a hierarchical meta-learning framework, this approach further enhances the model's training stability and feature modeling performance.

2)Contextual Temporal Interaction Relay (CIR): To mitigate the problem of information decay in long-term temporal modeling, we design a context-temporal interaction relay to assist Mamba's coherent modeling. This comprises two core components: a cyclic memory storage unit and an adaptive gated fusion unit. The cyclic memory storage unit is responsible for storing key global information corresponding to the previous $t-1$ EEG sub-time windows, including trend change markers and meta-learning adaptation parameters, providing historical context to support feature modeling for the current sub-time window and preventing the loss of long-term temporal information; The stored historical context information is denoted as CIR_{t-1} , and the update rule is shown in Equation (7):

$$CIR_{t-1} = \{F_{mamba}^{(t-1)}, \theta_{meta}^{(t-1)}, M^{(t-1)}\} \quad (7)$$

Here, $F_{mamba}^{(t-1)}$ represents the Mamba output features from the previous sub-time window, $\theta_{meta}^{(t-1)}$ represents the meta-learning adaptation parameters, and $M^{(t-1)}$ represents the trend change flag; together, these three elements constitute the historical context used for modeling at the current time step.

The adaptive gate fusion unit dynamically balances the contributions of the sub-time-window features from the current Mamba output and the historical memory information; the fusion formula is shown in Equation (8):

$$F_{global_raw} = \gamma \cdot F_{mamba} + (1 - \gamma) \cdot CIR_{t-1} \quad (8)$$

In particular, $\gamma = \sigma(W \cdot [F_{mamba}, CIR_{t-1}])$ represents the adaptive gating weight (where σ denotes the sigmoid function). The adaptive gate-controlled fusion unit adjusts the fusion ratio based on the emotional temporal patterns of the current sub-time window. When a key emotional response occurs, it increases the γ weight of the current

feature; conversely, it enhances the weight of historical memory to ensure the consistency and accuracy of long-term emotional trend modeling.

3)Global Meta-Learner: To accommodate the heterogeneity in the distribution of global features across different subjects—such as individual variations in the intensity of EEG frequency band responses—we reuse the parameterized structure of local meta-learners. Based on the initial global features output by Mamba and CIR, we adjust the parameters of both models to ultimately produce the sub-stream global features $F_{global} \in \mathbb{R}^{128}$, which have been individually adapted. The global meta-learner and the local meta-learner together form the hierarchical meta-learning framework referred to in the title of this paper; their collaborative operation enables cross-subject adaptation, which is key to enhancing the generalization capabilities of the HMABRF.

4)Local-Global Collaborative Alignment: To enhance the semantic consistency between F_{local} and F_{global} , we introduce a self-supervised local-global alignment loss. Using a two-layer fully connected network and a projection head in the BN layer, F_{local} and F_{global} are mapped to the same semantic space to maximize their cosine similarity, as shown in Equation (9):

$$Loss_{align} = -\frac{1}{B} \sum_{b=1}^B \frac{\langle \Phi(F_{local}^{(b)}), \Phi(F_{global}^{(b)}) \rangle}{\|\Phi(F_{local}^{(b)})\|_2 \cdot \|\Phi(F_{global}^{(b)})\|_2} \quad (9)$$

This loss function performs backpropagation on the parameters of the LTIM and GTCM models to ensure coordination between short-term local responses and long-term global trends, laying the groundwork for subsequent fusion among sub-streams.

3.4. Cross-Subflow Adaptive Aggregation and Emotion Classification

To integrate whole-brain features for accurate emotion recognition, we design a hierarchical adaptive fusion architecture that aligns with the logic of adaptive brain region segmentation. This architecture follows a progressive fusion logic—from sub-streams to macro-brain regions and across macro-brain regions—to ensure the gradual aggregation of feature information from the local to the global level. First, each adaptive brain region sub-stream fuses the local features F_{local} (output by LTIM) with the global features F_{global} (output by GTCM) through element-wise addition, yielding the complete sub-stream features $F_{sub} \in \mathbb{R}^{128}$. This ensures a deep coupling between the local short-term responses and the global long-term trends within the sub-stream. The fusion formula is shown below:

$$F_{sub} = F_{local} + F_{global} \quad (10)$$

We then adopt a hierarchical strategy of “fusion within macro-brain regions followed by cross-macro-brain region fusion”: First, using a multi-head self-attention mechanism (Cao, Dong and Chen, 2025), we assign weights to the features of all sub-streams belonging to the same macro-brain region and aggregate them into a macro-brain-region-level feature $F_{macro} \in \mathbb{R}^{256}$. The aggregation process can be expressed as:

$$F_{macro} = \sum_{i=1}^{N_s} \alpha_i \cdot F_{sub}^{(i)} \quad (11)$$

Here, N_s represents the number of sub-streams within the macro-brain region, and α_i denotes the self-attention weight corresponding to the features of the i -th sub-stream, which is adaptively learned by the multi-head self-attention mechanism. Then, repeat the adaptive attention fusion process on the three macro-level brain region features to generate the whole-brain composite feature $F_{total} \in \mathbb{R}^{512}$, using the following fusion formula:

$$F_{total} = \sum_{j=1}^3 \beta_j \cdot F_{macro}^{(j)} \quad (12)$$

Here, β_j represents the adaptive attention weight corresponding to the j -th macro-brain region feature. Finally, a lightweight classification head consisting of two fully connected layers and a Dropout layer maps F_{total} to the emotion category space, and a Softmax function is applied to produce a probability distribution. The total model loss is a weighted sum of the multi-class losses, as shown in Equation (13):

$$Loss_{total} = \lambda_1 Loss_{contrast} + \lambda_2 Loss_{align} + \lambda_3 Loss_{ce} \quad (13)$$

Specifically, $Loss_{contrast}$ is a self-supervised contrastive loss used in adaptive brain region segmentation, $Loss_{align}$ is a local-global alignment loss, and $Loss_{ce}$ is a multi-class cross-entropy loss. The weight coefficients of the aforementioned multi-objective loss function are optimized using grid search to ensure that each branch's loss contributes equally to the overall model training and to ensure training stability. The weight search range is set as follows: $\lambda_1 \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$, $\lambda_2 \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6\}$, $\lambda_3 \in \{0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$. The optimization process employs 5-fold cross-validation on the SEED dataset, using both the classification accuracy of each model and the stability of loss convergence as dual optimization criteria. After exhaustively exploring all weight combinations, the weight configuration that yields the best model performance and the most stable training is selected. The optimal weights were ultimately determined to be $\lambda_1=0.2$, $\lambda_2=0.3$, and $\lambda_3=0.5$. This configuration effectively balances the training objectives of self-supervised contrastive learning, local-global feature alignment, and supervised classification tasks, thereby enhancing the model's generalization ability and feature consistency.

4. EXPERIMENTS

4.1. Experimental Setup

We use the PyTorch framework to train and evaluate models on an Intel Xeon Gold 6248 CPU and an NVIDIA RTX 4090 GPU, using the SEED and SEED-IV datasets. The default optimizer is Adam, with a learning rate of 0.001 and 150 training epochs. The batch sizes for the SEED and SEED-IV datasets are set to 32 and 64, respectively. We evaluate the model using leave-one-out cross-validation, with the training and test sets drawn from different subjects, consistent with the standard evaluation paradigm in the field of EEG emotion recognition. This study uses accuracy as the primary evaluation metric, while also employing precision, recall, and F1-score as supplementary evaluation metrics to comprehensively assess the model's classification performance and generalization ability across different emotion categories.

4.2. Analysis of Experimental Results

4.2.1. Comparison of Accuracy with State-of-the-Art Methods

To validate the performance advantages of the HMABRF model in cross-subject EEG emotion recognition tasks, this study employed a leave-one-subject-out (LOSO) cross-validation paradigm on the SEED and SEED-IV datasets to compare the model against various baseline methods, including traditional machine learning, classical deep convolutional networks, graph neural networks, and EEG-specific Transformers. The experimental results are shown in Table 1. In terms of overall performance, HMABRF achieved the current best recognition accuracy on both datasets, fully demonstrating the effectiveness and state-of-the-art nature of the proposed model architecture.

On the SEED dataset, HMABRF achieved an average accuracy of $85.17\% \pm 6.83\%$, which is significantly higher than the $84.21\% \pm 6.05\%$ achieved by EEGMatch and the $84.86\% \pm 6.21\%$ achieved by AMDet. It also outperformed cutting-edge graph neural networks and spatio-temporal CNN models proposed in recent years, such as DGAT

Table 1

Comparison of Emotion Recognition Accuracy of Different Methods on the SEED and SEED-IV Datasets.

Methods	Accuracy (Mean $\pm$ SD, %)	
	SEED	SEED-IV
DeepConvNet[26]	81.37 $\pm$ 5.89	70.52 $\pm$ 8.11
EEGNet[27]	82.54 $\pm$ 6.12	72.18 $\pm$ 7.35
TSception[28]	83.85 $\pm$ 5.72	73.56 $\pm$ 6.98
DGAT[29]	84.01 $\pm$ 5.88	74.09 $\pm$ 7.01
EEGMatch[22]	84.21 $\pm$ 6.05	74.89 $\pm$ 7.12
AMDET[30]	84.86 $\pm$ 6.21	75.63 $\pm$ 7.19
HMABRF(Ours)	85.17 $\pm$ 6.83	76.32 $\pm$ 7.28

and TSception. These results indicate that, compared to GNN methods such as DGAT that rely solely on static brain region topology modeling, CNN models such as TSception that extract spatiotemporal features at a single scale, and the multi-dimensional attention Transformer architecture (AMDET) designed for EEG, HMABRF, by integrating physiological prior knowledge with data-driven adaptive brain region flow partitioning, combined with a local-global feature co-adaptation module based on hierarchical meta-learning, more accurately captures emotion-related spatiotemporal correlation features in EEG, effectively mitigating individual differences across subjects. On the SEED-IV dataset, HMABRF achieved an accuracy of 76.32% $\pm$ 7.28%, again ranking first among all compared methods. Although SEED-IV is a dataset with finer-grained emotional categories, resulting in lower feature distinctiveness and greater difficulty in generalizing across participants, HMABRF still maintains its performance advantage, further demonstrating the model's robustness and generalization ability. In contrast, early deep CNN baselines such as DeepConvNet and EEGNet performed significantly worse on both datasets compared to state-of-the-art GNN and Transformer-based deep models in recent years, reflecting that architectures relying solely on shallow spatiotemporal feature extraction struggle to effectively address the individual heterogeneity and complex spatiotemporal dynamics of EEG; While existing state-of-the-art methods have achieved performance improvements through attention mechanisms, dynamic graph modeling, or multi-scale feature fusion, they generally fail to strike a sufficient balance between physiological prior constraints of brain regions and data-driven flexibility. In fine-grained emotion recognition tasks such as SEED-IV, these methods still suffer from inadequate representation of emotion-related features and insufficient cross-subject generalization capabilities.

Overall, by employing adaptive brain region segmentation and a design that synergistically adapts local and global features, HMABRF effectively overcomes the performance bottlenecks of existing EEG emotion recognition methods. It not only outperforms mainstream Transformer and GNN methods on the SEED dataset but also maintains a

performance lead on the more challenging SEED-IV dataset, thereby providing an efficient and robust new solution for cross-subject EEG emotion recognition.

4.2.2. Comparison of Model Performance Based on Multidimensional Evaluation Metrics

The previous section compared the overall performance of our model against all baseline methods using accuracy. To more comprehensively evaluate the model's classification performance across different emotion categories, this section selects three representative mainstream methods—EEGNet, TSception, and AMDET—for a multi-dimensional metric comparison. This study introduces three metrics—precision, recall, and F1-score—to conduct a comparative analysis in terms of classification accuracy, identification completeness, and overall performance. The results are presented visually in Figure 2.

As shown in Figure 2, the HMABRF model proposed in this paper demonstrates superior overall performance and excellent generalization capabilities across various evaluation metrics in the emotion classification task. In the three-class emotion recognition task on the SEED dataset, HMABRF consistently outperformed all baseline models across the three core classification metrics—precision, recall, and F1 score—demonstrating the model's ability to accurately capture and effectively distinguish emotional features in EEG. On the SEED-IV dataset, which features finer-grained categories and greater recognition difficulty, HMABRF continues to maintain a consistent lead across all metrics, with less fluctuation in the performance of the three metrics. This indicates that, in multi-category, fine-grained sentiment recognition scenarios, the model demonstrates more balanced classification performance across different sentiment categories, exhibits no recognition bias toward specific categories, and possesses stronger cross-category adaptability. A comprehensive analysis of the multi-dimensional evaluation results reveals that HMABRF not only demonstrates superior overall recognition accuracy but also achieves synergistic optimization in classification precision, recall completeness, and overall stability. It effectively balances the core trade-off between precise discrimination and comprehensive coverage in emotion classification tasks, further validating the proposed method's effectiveness, robustness, and practical application potential.

4.2.3. Emotion Recognition Specificity and Visualization Analysis of EEG Physiological Features

To investigate the recognition specificity of HMABRF for different emotions and the physiological validity of the extracted features, we plotted confusion matrices for emotion recognition and EEG activation maps for the model on the SEED and SEED-IV datasets, respectively.

As shown in the confusion matrix in Figure 3, the proportion of emotional confusion across arousal levels in SEED is extremely low,

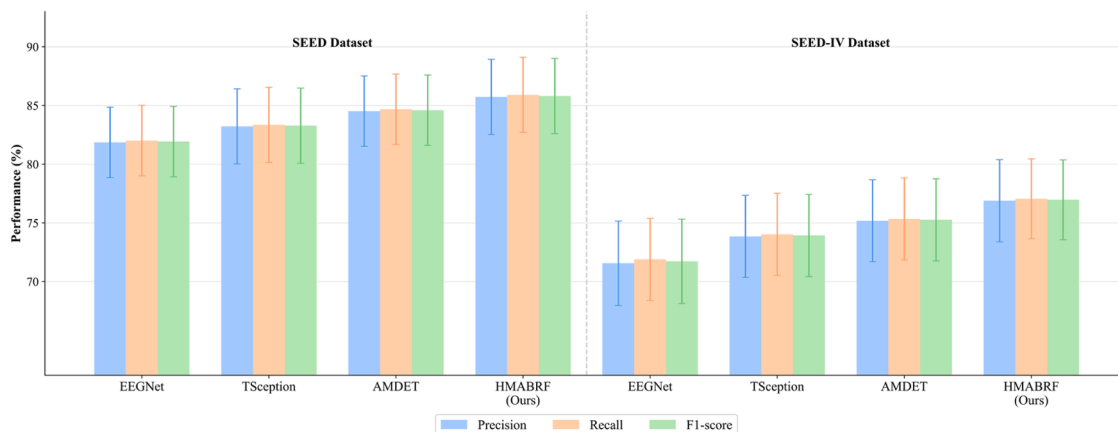


Fig. 2. Multi-dimensional Performance Comparison of Different Models.

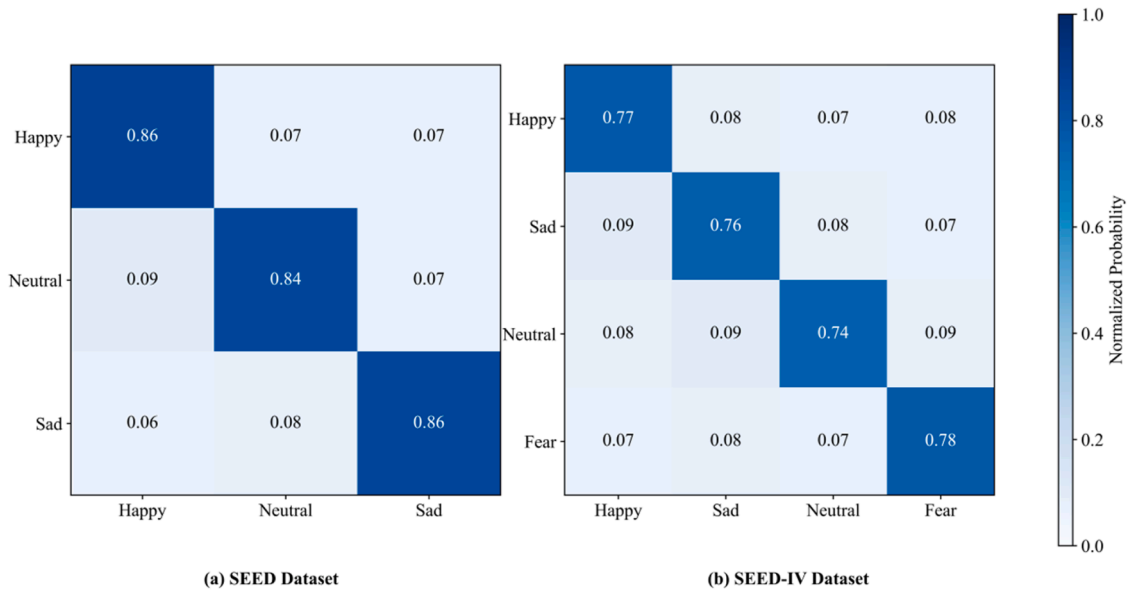


Fig. 3. Confusion matrices of emotion recognition by HMABRF on SEED and SEED-IV datasets.

indicating that the model accurately captures the distinct EEG patterns of high- and low-arousal emotions through adaptive brain region segmentation; In SEED-IV, confusion is concentrated among emotions of the same valence or similar arousal levels, further confirming that the features extracted by the model align with the neural processing logic of emotions. This consistent low rate of confusion not only reflects the model's targeted capture of core emotional features but also demonstrates its ability to generalize these features across different subjects.

Figure 4 shows that all emotions elicit significant activation in the prefrontal cortex, a region central to emotional processing and regulation. Sadness is characterized by distinct bilateral temporal lobe activation, consistent with the physiological role of the temporal lobes in encoding emotional memories. Neutral emotions do not show any specific regions of activation, and whole-brain EEG features exhibit low amplitude and uniform distribution. This demonstrates that the model is not merely a data-fitting exercise but captures emotion-related EEG features consistent with physiological a priori expectations.

4.2.4. Convergence of Model Training and Verification of Generalization Ability

To verify the stability of the HMABRF training process and its cross-subject generalization capability, we record the loss values and recognition accuracy for each training epoch on the SEED and SEED-IV datasets and plot training convergence curves. Figure 5(a) shows the results for the SEED dataset, while Figure 5(b) shows the results for the SEED-IV dataset.

As shown in the convergence curve in Figure 5, the training loss of HMABRF on both datasets decreases rapidly as the number of training iterations increases, leveling off around 80 iterations without any significant increase or fluctuation. This indicates that the model's network architecture, loss function design, and hyperparameter configuration ensure training stability and effectively prevent the vanishing or exploding gradient problem. At the same time, the model's training accuracy increases steadily with the number of training iterations and eventually stabilized, showing no obvious signs of overfitting. This laid a solid foundation for the model's excellent performance on the test sets, with the final model achieving cross-subject recognition accuracies of 85.17% and 76.32% on the SEED and SEED-IV test sets, respectively. The core reason why the model achieves stable convergence during training and excellent generalization performance during testing lies in the fact that self-supervised contrastive learning and local-global alignment loss enhance the semantic consistency and robustness of features; the LTIM and GTCM modules in the hierarchical meta-learning framework dynamically adapt to individual differences, reducing the heterogeneity of feature distributions across subjects; lightweight designs such as deep separable convolutions and residual connections also effectively suppress overfitting, enabling the features learned by the model to possess strong cross-subject generalization capabilities.

4.3. Ablation experiments

To validate the effectiveness of each core component of HMABRF, we

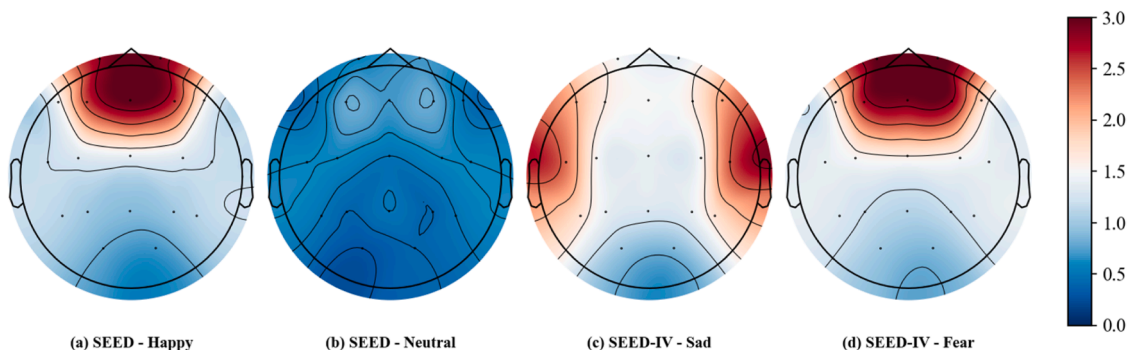


Fig. 4. EEG activation topographic maps corresponding to different emotions recognized by HMABRF.

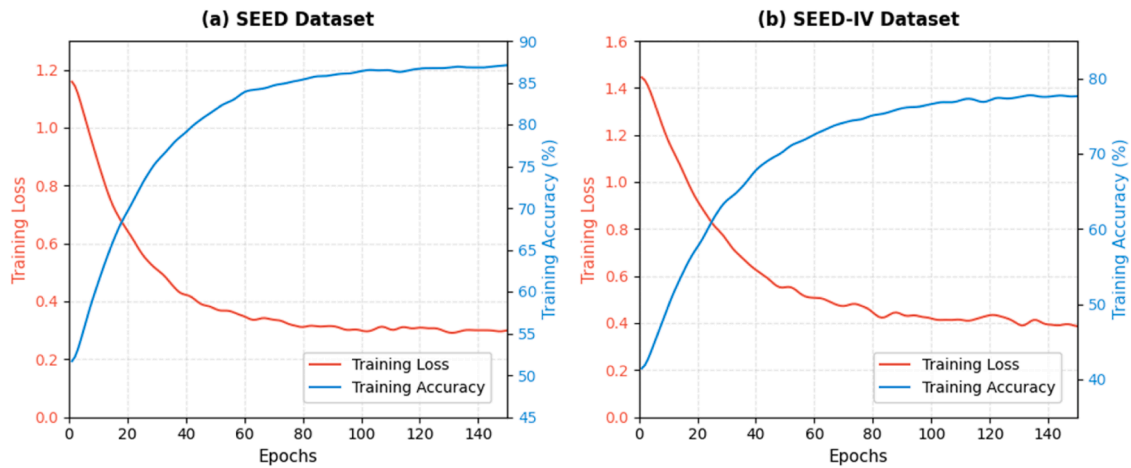


Fig. 5. Training Convergence Curves of HMABRF on the SEED and SEED-IV Datasets.

designed ablation experiments on the SEED and SEED-IV datasets; the results are shown in Table 2.

The results in Table 2 show that self-supervised contrastive learning, adaptive brain region flow partitioning, LTIM, GTCM, and the local-global alignment all significantly improve model performance. In particular, when the self-supervised contrastive learning was removed and the model was trained using supervised learning alone, its performance declined significantly. This confirms that self-supervised contrastive learning enhances the model's ability to extract features from unlabeled EEG data, thereby effectively improving the model's generalization and feature robustness; After removing the ABRF and replacing it with a fixed anatomical brain region segmentation, the model's performance declined further, directly validating the core contribution of adaptive dynamic clustering in brain region segmentation. Compared to fixed anatomical brain regions, the adaptive strategy can optimize sub-stream segmentation by incorporating individual EEG data features, better accounting for individual differences in emotional neural activity, and thereby enhancing feature representation capabilities; After removing the LTIM and replacing it with a standard convolutional feature interaction layer, the model's performance declined significantly, providing intuitive validation of LTIM's (including BIE) ability to accurately capture and synergistically enhance local multiscale EEG features; Furthermore, the significant drop in model performance following the replacement of GTCM with BiLSTM provides intuitive validation of the core value of the Mamba global temporal modeling mechanism in GTCM; After removing the hierarchical meta-learner, the model's performance declined significantly, directly confirming the core value of the meta-learning-driven individual adaptation mechanism. As a key pillar of LTIM and GTCM, hierarchical meta-learning dynamically adapts to differences in local and global feature distributions across different subjects, enabling the model to update parameters in a personalized manner across subjects. This effectively mitigates the

performance degradation of population-trained models during cross-subject testing and is a critical component for improving cross-subject generalization performance. Self-supervised contrastive learning enables the effective extraction of unlabeled features; ABRF establishes the foundation for brain region feature extraction through a dynamic balance between physiological prior knowledge and data-driven insights; LTIM enhances the precise capture and adaptation of individual local multi-scale features; GTCM optimizes the modeling consistency of global long-term emotional trends; and Loss_align ensures semantic coherence between local short-term responses and global long-term features; The optimal performance of the complete model fully validates the rationality of the method's design and the synergistic and complementary value of each module.

To further validate the appropriateness of the number of experts in the LTIM module, we conducted ablation experiments comparing different numbers of experts; the results are shown in Table 3.

As shown in Table 3, as the number of experts increases from 2 to 4, the model's recognition accuracy continues to improve; however, once the number exceeds 4, accuracy no longer increases and may even decline slightly, while the number of model parameters and computational overhead increase significantly. These results confirm that four experts represent the optimal configuration for balancing feature extraction capability, model efficiency, and recognition performance, and also validate the design logic described earlier regarding the four core local feature dimensions of EEG.

5. Conclusion

This paper addresses the core challenges in EEG emotion recognition—namely, the difficulty of balancing physiological a priori knowledge with data-driven approaches, and the lack of coordination between individual adaptation and the integration of local and global features—by proposing a novel EEG emotion recognition method called HMABRF. This method balances physiological logic with data flexibility through adaptive brain region segmentation. It relies on the collaborative adaptation of the LTIM and GTCM dual modules—driven by

Table 2
Ablation Experiment Results of Each Core Component of HMABRF.

Model variant	SEED Accuracy (Mean±SD, %)	SEED-IV Accuracy (Mean±SD, %)
No self-supervised contrastive learning (only supervised training)	82.85±6.90	73.75±7.38
No ABRF (replaced by fixed anatomical regions)	81.25±6.97	72.11±7.56
No LTIM (replaced by standard convolution)	82.03±6.89	73.04±7.41
No GTCM (replaced by BiLSTM)	80.76±7.01	71.88±7.62
No Meta-Learner	82.41±7.05	73.36±7.44
No Loss_align	83.24±6.91	74.27±7.35
HMABRF	85.17±6.83	76.32±7.28

Table 3
Effect of the Number of Experts on Model Performance.

Number of experts	SEED Accuracy (Mean±SD, %)	SEED-IV Accuracy (Mean±SD, %)
2	82.36±6.94	73.18±7.40
3	83.71±6.85	74.92±7.33
4	85.17±6.83	76.32±7.28
5	84.93±6.86	76.15±7.31
6	84.78±6.90	75.93±7.36

hierarchical meta-learning—to account for individual differences in local and global features, while combining self-supervised alignment loss to enhance feature consistency. The experimental results demonstrate that HMABRF outperforms existing state-of-the-art methods on both the SEED and SEED-IV datasets, and ablation experiments validate the necessity and synergistic value of each core component. This method provides an improved feature extraction and individual adaptation scheme for EEG emotion recognition and shows great potential for practical application. **Future research will focus on multimodal and complex emotion recognition tasks.**

CRedit authorship contribution statement

Hao Liu: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Conceptualization. **Yindong Dong:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **Yiming Tang:** Writing – review & editing, Visualization, Validation. **Jingjing Deng:** Writing – review & editing, Methodology, Formal analysis. **Yong Ye:** Writing – review & editing, Validation, Conceptualization. **Yubo Xing:** Writing – review & editing, Visualization, Methodology. **Xiule Wang:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors have declared that no competing interests exist.

Acknowledgements

This work was supported in part by the National Natural Science Foundation of China No. 62576130, the Key Project of Humanities and Social Sciences Research of Anhui Provincial Department of Education in Colleges and Universities No.2023AH050966, and the Anhui Provincial College Students' Innovation and Entrepreneurship Training Program (Provincial Level) under Grant No. S202510364016.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijcce.2026.04.005](https://doi.org/10.1016/j.ijcce.2026.04.005).

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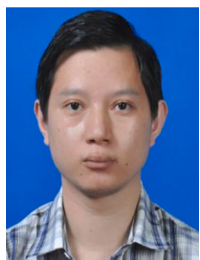
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Hao Liu is a student majoring in Computer Science and Technology at the School of Artificial Intelligence, Anhui Agricultural University. Throughout his academic journey, he has shown a passion for research and innovation in the field of computer science.



Yindong Dong received Ph.D. degree in computer application technology from Hefei University of Technology, China in 2021. He works for the School of Artificial Intelligence, Anhui Agricultural University since June 2021. His research interests are affective computing, natural language processing, and intelligent recognition based on physiological signals.



Yiming Tang received his Ph.D. degree in Computer Application Technology from Hefei University of Technology in Hefei, China, in 2011. He is currently a Professor at the School of Computer Science and Information Engineering of the university. His research areas cover Artificial Intelligence, Affective Computing, Fuzzy Systems, Image Processing, and Collaborative Design.



Jingjing Deng received her Master's degree in Modern and Contemporary Chinese Literature from Anhui University in Hefei, China, in 2017. Currently, she serves as a counselor and lecturer at the School of Artificial Intelligence, Anhui Agricultural University, and her research focuses on mental health education for college students.



Yong Ye is a professor, master's supervisor, and the program leader of Logistics Engineering at the School of Artificial Intelligence, Anhui Agricultural University. He was once a visiting scholar at Mie University in Japan. He focuses on research in agricultural logistics, supply chain, and logistics system simulation.



Yubo Xing is a student majoring in Computer Science and Technology at the School of Artificial Intelligence, Anhui Agricultural University.



Xiule Wang is an undergraduate student in Computer Science and Technology at Anhui Agricultural University. His research interests focus on artificial intelligence and machine learning.